

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\left(u_{\underline{m}}^{\underline{a}}\right)=\left(u_{\underline{m}}^a,u_{\underline{m}}^i\right)\in SO(1,D-1)\Leftrightarrow u_{\underline{m}}^{\underline{a}}\eta^{mn}u_{\underline{n}}^{\underline{b}}=\eta^{ab}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\mathcal{A}=\mathcal{A}_0+\mathcal{A}^{\text{int}},\quad \mathcal{A}_0=\int_{t_a}^{t_b}dt\left[\frac{M}{2}\dot{x}^2-\frac{\Omega^2}{2}x^2\right],\quad \mathcal{A}^{\text{int}}=\int_{t_a}^{t_b}dtV_{\Omega}^{\text{int}}\equiv\int_{t_a}^{t_b}dt\left[-\frac{(\omega^2-\Omega^2)}{2}x^2-V^{\text{int}}(x)\right],$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\mathbf{Z}(\mathbf{x})=\frac{1}{\mathbf{x}}+\sum_{q\neq 0}\{\frac{1}{\mathbf{x}-\mathbf{q}}+\frac{1}{\mathbf{q}}+\frac{1}{\mathbf{q}}\mathbf{x}\frac{1}{\mathbf{q}}+\frac{1}{\mathbf{q}}\mathbf{x}\frac{1}{\mathbf{q}}\mathbf{x}\frac{1}{\mathbf{q}}+\frac{1}{\mathbf{q}}\mathbf{x}\frac{1}{\mathbf{q}}\mathbf{x}\frac{1}{\mathbf{q}}\mathbf{x}\frac{1}{\mathbf{q}}\},$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$1=\int d^2\mathbf{r}\,|\Psi(\mathbf{r})|^2=A_l^2\,2\pi\kappa^{-2}\left\{\mathcal{K}_{i\Theta}(\kappa a)+\left(\frac{\kappa}{\tilde{k}}\right)^2\left[\frac{K_{i\Theta}(\kappa a)}{J_l(\tilde{k}a)}\right]^2\mathcal{J}_l(\tilde{k}a)\right\},$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\dot{\epsilon}_{\delta\sigma}=\epsilon_{\delta\sigma}\left[3\frac{\dot{\bar{a}}_R}{\bar{a}_R}(1+w)+2\frac{\dot{\bar{c}}_4}{\bar{c}_4}\frac{1+\epsilon_{\delta\sigma}}{1+\epsilon}+\left(\frac{\dot{\bar{G}}_4}{\bar{G}_4}-4\frac{\dot{\bar{c}}_4}{\bar{c}_4}\right)(1+\epsilon_{\delta\sigma})\right].$$